

MADHUSUDAN BHARDWAJ

+918287031247 | madhusudanbhardwaj8287@gmail.com | [LinkedIn](#) | [portfolio](#)

EDUCATION

- Netaji Subhas University of Technology

Masters of Technology in Artificial Intelligence

August 2024 – Present

Delhi, India

- Krishna Engineering College (affiliated to AKTU)

Bachelor of Technology in Computer Science and Engineering

July 2020 – June 2024

Ghaziabad, India

INTERNSHIP

- NimbusGaming

March 2024 – June 2024

- Worked on developing an **AI-powered Retrieval-Augmented Generation (RAG) chatbot** using **Generative AI**, **semantic search**, and **Large Language Models (LLMs)** to deliver accurate and context-aware responses.
- Implemented **vector embeddings**, **similarity search**, and **LLM integration pipelines** to improve response relevance, retrieval accuracy, and overall system performance.
- Designed a **context-aware query handling and information retrieval pipeline** for efficient document understanding and intelligent response generation.

PROJECTS

- Openleaf

- Developed an **Agentic AI-based Research Paper Writer** using **Large Language Models (LLMs)** and **Human-in-the-Loop (HITL)** workflows for automated research draft generation.
- Implemented **MCP-based context management**, **multi-step reasoning**, and **long-form content generation** to improve coherence and research accuracy.
- Built an automated **NLP pipeline** for **topic analysis**, **literature review synthesis**, **prompt engineering**, and structured document generation.
- Tools:** Python, LangChain, LangGraph, LLMs, Prompt Engineering, MCP.

- SmartRAG

- Developed an **AI-powered Retrieval-Augmented Generation (RAG) chatbot** using **BM25**, **FAISS**, and **Large Language Models (LLMs)** for intelligent document-based question answering.
- Implemented **semantic search**, **vector embeddings**, and **context-aware response generation** to improve retrieval accuracy and efficient information extraction from PDF/documents.
- Tools:** Python, LangChain, FAISS, BM25, Hugging Face, LLMs.

- DoctorG

- Developed a **Multi-Layer Artificial Neural Network (ANN)** model for **multi-class disease prediction** to identify the **top-5 probable diseases** from user symptoms using weighted features and embeddings.
- Built an **NLP-based preprocessing pipeline** including **text normalization**, **feature engineering**, **vectorization**, and **embeddings** to improve model accuracy and generalization.
- Implemented **Softmax-based probability ranking** and integrated **Generative AI** for confidence-driven predictions and automated disease description generation.
- Tools:** Python, TensorFlow/Keras, Scikit-learn, Pandas, NumPy, NLP, Generative AI.

TECHNICAL SKILLS

- Languages:** Python, Java, C/C++, JavaScript
- Libraries/Frameworks:** TensorFlow, Keras, OpenCV, Scikit-learn
- Concepts:** Machine Learning, Deep Learning, NLP, Computer Vision, Data Structures & Algorithms, MCP
- Tools:** Git, VS Code, Jupyter Notebook, Google Colab
- Databases:** MySQL

ACHIEVEMENTS

- Solved over 700+ DSA problems on [Leetcode](#), [CodingNinja](#) demonstrating strong problem-solving skills.
- Certificates**